

Stéphane Boyera

ICT for Development Expert

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Career Objectives

I'm passionate about Web and mobile technologies innovation, in particular, innovation related to social and economic development: providing the benefits of ICTs and mobile ICTs, with emphasis upon marginalized communities and people who are currently excluded from the Information Society and that cannot access adequate services such as education, health, livelihood, or cannot participate and be informed about the democratic life of their country. In that regards, I'm also passionate about the Open Data, Open Governance movement and Open Government Data initiatives all over the World and particularly in low and middle-income countries.

Areas of Expertise

- ICT for Development (mobile, voice, Web and radio): Strategy, project management, field studies, development and deployment, business modeling, sustainability and scalability. Sector expertise: Agriculture, Media and Education
- Web and mobile technologies – open standards and open source
- Multi-stakeholder partnership development: North/South, international organizations, corporate, NGO/civil society organizations, academic
- Technology and mobile entrepreneurship in developing countries
- Open Data: Strategy, project management, feasibility studies, implementation.
- Open governance and open government data (OGD) in developing countries: Strategy, project management, feasibility studies, implementation.

Work Experience

SBC4D ***January 2009 to Present***

Where ICT meets Social and Economic Development

Toulouse, France – http://www.sbc4d.com

Position: **Founder and Chief Executive Officer**

Consulting on ICT for Development (ICTD), mobile technologies, mobile technologies for development, mobile entrepreneurship, ICTD business modeling, data collection process and tools, open government data, software development, multi-stakeholders project management, Web standards, Web technologies, cloud computing, scientific and technical reviews, application of advanced technologies to improve productivity, or social impact.

Principal clients include non-profits organizations (NGOs and foundations such as the MTV Staying Alive Foundation, Web Foundation – see below for details), Development agencies such as AFD (Agence Francaise de Development - education, mobile services), international organizations such as W3C (mobile money, Web payments), The World Bank (OGD feasibility studies), CTA (agriculture) or AUF (Agence Universitaire de la Francophonie - education, mobile services), different UN agencies such as IFAD (open data) or International Trade Center (agriculture, ICTD strategy), governments, corporates such as Orange (research studies on e.g. mobile data collection), or Comviva (mobile money), and startups such as e.g. Akengo (business models, mobile technologies).

World Wide Web Foundation ***January 2009 to June 2013***

A foundation dedicated to driving high-impact initiatives to empower disadvantaged people through the Web.
Geneva, Switzerland - http://www.webfoundation.org

Position: **Lead Program Manager**

I helped setup the Web Foundation and was in charge of defining the Web Foundation work program, identifying projects in the field (Sub-Saharan Africa, Asia and South America), developing local and global partnership, managing research projects, implementation projects, and studies. This position includes being a member of the Web Foundation management team. Major achievements:

- Designed and launched the program strategy around 5 themes: Voice Technologies, Web for Agriculture, Open Government Data, Mobile Entrepreneurship, and Web Index
- Setup and recruited the project team (8 people) including international experts in related domains.
- Developed an international network of partners from all categories of stakeholders: International organizations (The World Bank, UNESCO, ICANN, IGF, etc.), corporate (Orange, Vodafone, Nuance, Al Jazeera, etc.), academics (VU Amsterdam, TNO, CRS4, CSIR/Meraka Institute – South Africa, North West University- South Africa, University Cheick Anta Diop – Senegal, Gaston Berger University – Senegal, ESMT –Senegal, etc.), NGO and civil society organizations (Sahel Eco-Mali, Reseau MARP – Burkina, One World South Asia – India, iHub – Kenya, etc.), donors (AFD, IDRC, USAID, Hewlett Foundation, Rockefeller Foundation, etc.)
- Managed the technical aspect of the MyWorld project, a UN initiative aiming at running a worldwide survey to ask people their input about post-2015 MDGs. This includes organizing a mobile survey in 20+ countries
- Part of the founding team and member of the Web Alliance for Regreening in Africa (W4RA)
- Launched mobile training lab in Ghana, Kenya and Senegal
- Deployed voice-based projects for farmers and radio stations in Mali, Ghana, Burkina Faso and India
- Managed the OGD initiative of the government of Ghana
- Developed, and successfully managed the EU-funded VOICES project with 13 partners (industry, academics, NGOs) from 8 countries (3 in Africa), with pilots in Mali and Senegal (4M USD total budget)
- Obtained international recognition for these efforts through publications, keynotes at important conferences (e.g. UNESCO 1st Mobile Learning Week), talks and press appearances. See Annex for publication details.

World Wide Web Consortium (W3C) ***June 1995 to September 2009***

A consortium of leading organizations developing open Web standards and directed by Tim Berners-Lee, inventor of the Web.
Sophia-Antipolis, France – http://www.w3.org

Position: **Chair of Mobile Web for Social Development group** *(January 2006 - September 2009)*

I was in charge of defining a new W3C work program about leveraging the Web in developing countries. This included the design of the W3C roadmap, the development of international partnership, the development of an outreach strategy to establish W3C as a major player in the domain, and fundraising to support the activity. Major achievements:

- Developed and released the W3C Mobile Web for Social Development Roadmap
- Developed, submitted, successfully managed and completed the EU-Funded Digital World Forum project including 8 partners from 5 countries – 3 African ones.
- Obtained international recognition for these efforts through scientific paper publications, talks and press appearances (e.g. featured in The Economist - September 2008). See Annex for publication details.

My work in this position was a major contribution to the design, and launch of the World Wide Web Foundation.

Position: **Ubiquitous Web Domain co-manager** *(January 2004 - December 2005)*

I was managing part of the Ubiquitous Web Domain. This includes overviewing mobile activities of the domain (Mobile Web Initiative, Voice activity, Multimodal Activity) and managing related activity leaders (5 people). It also includes fundraising (project proposal development) and proposal management. Major achievements:

- Co-founded the W3C Mobile Web Initiative, the first international mobile industry-wide initiative for Web access from mobile phones. This included all the major mobile players (from Vodafone to Orange, T-Mobile, O2, Airtel, to Nokia, Eriksson to Opera, Mozilla, Microsoft, etc.)
- Managed several dimensions of different EU-funded projects (3GWeb Project, MobiWeb 2.0 Project)
- Managed and launched the W3C online training activity
- Obtained international recognition for these efforts through scientific paper publications, talks and press appearances. See Annex for publication details.

Position: **Device Independence Activity Lead** *(January 2000 - December 2003)*
I was in charge of the management of the Device Independence activity. This includes leading corresponding working groups, participating in the development of related standards, developing W3C membership in the area, and managing communication and outreach about related standards. Major achievements:

- Contributed to W3C standards: CC/PP Structure and Vocabularies, DIAL : Device Independent Authoring Language and DI-Select
- Obtained international recognition for these efforts through scientific paper publications, talks and press appearances. See Annex for publication details.

Position: **W3C-EU System Team Leader** *(June 1995 – December 1999)*
I was in charge of setting-up the system infrastructure of the W3C EU team, as well as operating it, providing support to the team, setting-up collaboration tools and infrastructure for the EU team, and for the cross-collaboration between the EU-US-Japan teams as well as other international offices of W3C. Major Achievements include:

- Developed and deployed the W3C web mirroring system across 7 countries in Europe and North Africa using satellite communication
- Organized technical infrastructure and support of WWW95 conference with 1200+ participants

INRIA ***September 1993 to June 1995***
The French national research institute on computer science and automation
***Sophia-Antipolis, France,* - http://www.inria.fr**

Position**: Research Engineer**
As part of the GENIE project (joint initiative between INRIA and Dassault Aviation), I was in charge of identifying and characterizing multi-expertise interactions and synergies. Research themes include:

- Knowledge acquisition and representation in a multi-expertise environment
- Artificial intelligence, Conceptual graphs and ontologies

These research activities led to a number of scientific publications. See Annex for publication details.
The position also included teaching activities at Msc (artificial intelligence, knowledge acquisition) and undergratudate levels (mathematics and computer science).

References are available on request.

Education

ESSI, University of Sophia-Antipolis, France - *September 1992 to August 1993*
Master of Advanced Studies Diploma (DEA) in Computer Science (software development, networking & telecommunication) received in September 1993

ESSTIN, Sophia-Antipolis, France - *September 1988 to July 1993*
Master of Science in Engineering & Telecommunication. Degree received in July 1993.

Skills

Organizational
Leadership skills: I successfully managed international teams of 2 to 10 people in Web and mobile projects; defined, started and managed a complex, EU-Africa project involving multiple partners from Government to NGO to corporate to academics. Financial skills: extensive fund-raising and budget management experience.

Technical
I have 19+ years of experience in Web technologies and open standards; 10+ years of experience in mobile technologies, in Web access from mobile and in voice technology; 8+ years of experience in ICT for development, ICT usage by marginalized communities in rural areas of developing countries, and how to mix traditional media (radio) and ICT; 4+ years of experience in Open Government Data and related technologies.

Social
I have 19+ years of experience in working in an international setting. I was involved during 14+ years in the development of Web standards based on consensus building, requiring leading 40+ person groups without hierarchical authority. I've been working in more than 32 developing countries (Africa, Latin America, South and South-East Asia). I'm productive and reliable either as an independent agent or team member. I am comfortable in both non-profit and for-profit organizations and environments. I'm creative, persistent and adaptable and have excellent verbal and written communication skills.

Languages

French

Mother tongue

English

Fluent in speaking, reading, writing and understanding

Spanish

Basic level

Japanese, Indonesian

Notions

Resume updated on December 2013

Annex: Selection of scientific publications, talks and press appearance

2013

- Co-Laureate of the 100 Innovations for Africa selection by the French Ministry of Foreign Affairs (see a presentation of the project), Paris, France, 5 December 2013.
- "The Web of Voices", S. Boyera, session organizer at the European Development Days 2013, 26 November 2013, Brussels, Belgium.
- "Business matching service for small-holder illiterate farmers", S. Boyera, Invited Demonstration at the 1st ICT4ag conference, Kigali, Rwanda, 4 November 2013
- "Semantic Web for Social & Economic Development?", Stephane Boyera, Principale Keynote, Downscale 2013, 19 September 2013, Geneva Switzerland
- "Indonesia OGD Readiness Assessment study", Jose M. Alonso, Stephane Boyera, Aman Grewal, Carlos Iglesias, Andreas Pawelke, June 2013, http://www.webfoundation.org/wp-content/uploads/2013/06/OGD-Indonesia-FINAL-for-publication.pdf
- "Voices Gives Illiterate Access to the Internet", New Design, 7 May 2013
- "Open Data Readiness Assessment Tool", April 2013, French Translation, Stephane Boyera, Axel Rifon, Christelle Signo Kouame http://personal.crocodoc.com/XYJd9GT,
- "Voice-based Web access helps illiterate get online", New Scientist, 25 April 2013
- "Introduction to ICT for social and economic development", Stéphane Boyera, Principal Keynote, Voice technologies workshop, 23 April 2013, Bamako, Mali
- "Les services de collecte de données par mobile en Afrique sub-saharienne", Kitty Boullé, Stéphane Boyera & Maxime Froumentin, Colloque International « Localisation et Réinvention de la téléphonie mobile », April 2013, Cotonou, Bénin
- "Foroba Blon, a citizen journalism platform for rural Mali", Stéphane Boyera, Voice technologies workshop, 23 April 2013, Bamako, Mali
- "Voice-based Web access in rural Africa" by Nana Baah Gyan, Victor de Boer, Anna Bon, Chris van Aart, Stéphane Boyera, Hans Akkermans, Mary Allen, Aman Grawal and Max Froumentin. Accepted for publication at ACM Web Science 2013 conference (WebSci2013), May 2-4 2013, Paris, France. Nominated as part of the 8 best papers.
- "Use case and requirements analysis in a remote rural context in Mali" by Anna Bon, Victor de Boer, Nana Baah Gyan, Chris van Aart, Pieter De Leenheer, Wendelien Tuyp, Stéphane Boyera, Max Froumentin, Aman Grewal, Mary Allen, Amadou Tangara, Hans Akkermans. Accepted for publication at the 19th International Working Conference on Requirements Engineering: Foundation for Software Quality (REFSQ 2013), April8-11 2013, Essen, Germany
- "Voice Technologies for Social Development", Principal Keynote, Voice Technologies Workshop, 5-6 March, CSIR, South Africa
- "Les TIC pour le Développement: une approche multi-domaines", Keynote, February 7, IMAG, Grenoble, France
- "Mobile Data Collection Opportunities in Sub-Saharan Africa", study and reports, 14 Janvier 2013, http://public.webfoundation.org/2013/data-collection/findings.pdf

2012

- "Radio Marché: Distributed Voice-enabled Web Interfaced Market Information System under Rural Conditions" by Victor de Boer, Pieter De Leenheer, Anna Bon, Nana Baah Gyan, Chris van Aart, Christophe Guéret, Wendelien Tuyp, Stéphane Boyera, Mary Allen, Hans Akkermans. Accepted for publication in Proceedings of 24th International Conference on Advanced Information Systems Engineering, Caise2012, Gdansk, Poland, 25 29 June 2012.
- Principal Keynote at Spoken Language Technology for Under-resourced languages (SLTU) 2012
- "Bringing the Web of Data to Developing Countries - Linked Market Data in the Sahel Region". Poster by Victor de Boer, Nana Baah Gyan, Pieter de Leenheer, Anna Bon, Chris van Aart, Christophe Guéret, Wendelien Tuijp, Stéphane Boyera, Mary Allen, Hans Akkermans presented at Extended Semantic Web Conference 2012, Heraklion, Greece. **Award winning "best poster"**

- "The Web of Radios - Introducing African Community Radio as an interface to the Web of Data" by Anna Bon, Victor de Boer, Pieter De Leenheer, Chris van Aart, Nana Baah Gyan, Max Froumentin, Stéphane Boyera, Mary Allen, Hans Akkermans. Presented at the First International Workshop on DownScaling the Semantic Web, hosted by Extended Semantic Web Conference 2012.

2011

- **"**Technologies, Education and Socio-Economic Development**"** Principal Keynote at the UNESCO 1st mLearning Week conference

- Mobile and Web technologies in Social and Economic Development, Stéphane Boyera, Chair, Workshop, Dar Es Salam, Tanzania, June 2011
- "Making Broadband Accessible for All", Member of the expert panel, Vodafone Policy Paper Series - Number 12 - May 2011,
- "Ghana OGD Feasibility study", May 2011, Jose M. Alonso, Stephane Boyera, Aman Grewal, Carlos Iglesias, Steve Bratt, http://public.webfoundation.org/2011/05/OGD_Ghana.pdf
- "Chile OGD Feasibility study", May 2011, Jose M. Alonso, Stephane Boyera, Aman Grewal, Carlos de la Fuente, Alfonso García-Jove, Carlos Iglesias, Steve Bratt, http://public.webfoundation.org/2011/05/OGD_Chile.pdf
- "The Web of Voices: how to connect 4.5 billion internet-less people to the Web" by Chris van Aart, Anna Bon, Hans Akkermans, Victor de Boer, Stéphane Boyera, Wendelien Tuyp, Nana Baah Gyan: Outrageous Ideas, International Semantic Web Conference 2011 (ISWC 2011). **Public award winning paper. 3rd Place**.
- "VOIce-based Community-cEntric mobile Services for social development (VOICES)", by Stéphane Boyera. IST-Africa, Gaborone, Botswana.
- "Is (Web) Science ready for Empowerment" by Hans Akkermans, Nana Baah Gyan, Anna Bon, Wendelien Tuyp, Stéphane Boyera, Aman Grewal, Mary Allen. ACM Web Science Conference 2011.
- "Speech for the Developing World" by Stéphane Boyera. SpeechTek 2011, New-York, USA.

2010

- "Mobile Technologies for Social Development", Keynote at AFNOG Web workshop, Kigali, Rwanda.
- "Mobile Technologies for Social Development", Principal Keynote at the India National Forum on Mobile Applications for Inclusive Growth and Sustainable Development. April 2010, New-Delhi, India
- Member of the World Bank Expert Panel on Mobile for Rural Development
- Member of the World Bank Expert Panel on Mobile Health

2009

- "Empowering People", Stéphane Boyera, TedX Kampala, Uganda, November 2009
- "Mobile Web For Social Development Roadmap", by Stéphane Boyera, W3C Working Group Note, November 2009
- "Empowering people: The endeavor of the World Wide Web Foundation", Stéphane Boyera & Steve Bratt, Dies Natalis Symposium, Amsterdam, Netherlands, October 2009
- "The future of the Web is in Africa", Stéphane Boyera, 20th Anniversary of the Web:, Geneva, Switzerland
- Chair of the W3C Workshop on the Role of Mobile Technologies in fostering Social Development - Africa Perspective, Stéphane Boyera, Chair, Maputo, Mozambique, April 2009
- Mobile Phones in ICT4D, Stéphane Boyera, Development and Cooperation 2.0, Gijon, Spain, February 2009

2008

- "Using Web Technologies and mobile phones for Social Development: W3C Approach", paper by Stephane Boyera, 1st International Conference on M4D, Karlstadt, Sweden, December 2008

- "Technology, a platform for Development", Stéphane Boyera, Chatham House, London, UK, October 2008
- "The Meek shall inherit the Web", The Economist, September 2008
- "How mobile public services can transform lives", Nokia Expanding Horizon Q3 2008, September 2008
- "W3C Mobile Web for Social Development Program" , Stéphane Boyera, Mlife 2008, Antalya, Turkey, September 2008,
- "Digital World Forum: Low-cost technologies against the Digital Divide", Stéphane Boyera, 5th EuroAfrica-ICT Concertation Meeting, Brussels, Belgium, June 2008,

- "Mobile Web in Rural China?", Stéphane Boyera, WWW2008, Beijing, China, April 2008,
- "Using Mobile Web technologies in Social Development", Stéphane Boyera, HFT 2008, Kuala Lumpur, Malaysia, March 2008
- "Using the Mobile Web in Development", Stéphane Boyera, Workshop of DIS 2008, Cape Town, South Africa, February 2008
- "W3C Workshop on the Role of Mobile Technologies in fostering Social Development", Stéphane Boyera, Chair, Sao Paulo, Brazil, February 2008

2007

- "Potential and challenges of mobile internet services in Africa", Stéphane Boyera, 3GSM Africa, Cape Town, South Africa, November 2007,
- "Mobile and Governance", Stéphane Boyera & Jose Manuel Alonso, Interview by the World Bank, mGovernance Conference, The World Bank, November 2007
- "Leveraging Web access for rural communities and under-privileged populations", Stéphane Boyera, 5th Latin America Web Congress, Santiago de Chile, Chile, October 2007,
- "Opportunities and Challenges of Web Technologies on Mobile Platform", Stéphane Boyera, Datamatix Gitex Conference 2007, Dubai, UAE, September 2007,
- "Can the Mobile Web Bridge the Digital Divide?", Stéphane Boyera, ACM Interaction Magazine, Volume 14 , Issue 3 (May + June 2007)
- "The Mobile to Bridge the Digital Divide", Stéphane Boyera, 3GSM - East and Central Africa, Nairobi, Kenya, May 2007,
- "The W3C Mobile Web Initiative", Stéphane Boyera, W3C Southern Office Opening, Pretoria, South Africa, May 2007,
- "The Mobile Web to bridge the Digital Divide" (Paper - Presentation), Stéphane Boyera, IST-Africa 2007, Maputo, Mozambique, May 2007
- "1st Workshop on Mobile Internet User Experience", member of the program committee.

2006

- "Mobile Web in the Developing Countries", Stéphane Boyera, Workshop Chair, Bangalore, India, December 2006
- "Bridging the Digital Divide", Stéphane Boyera, MWI Seminar, Paris, November 2006

2005

- "W3C Mobile Web Initiative" and "The Device Independent Web", Stéphane Boyera, W3C India Office Opening, New Delhi, India, November 2005
- "W3C Mobile Web Initiative", Stéphane Boyera, WWW 2005, Chiba, Japan

2004

- "Workshop on Metadata for Content Adaptation", Stéphane Boyera, Co-Chair, Dublin, Ireland, October 2004
- "Device Independent Web", Stéphane Boyera, DIWE'04, Munich, Germany, July 2004
- "Web Mobile et Diversification des Terminaux", Stéphane Boyera, UBbiMob 2004, Sophia-Antipolis, June 2004 (french)
- "Device Independent Web", Stéphane Boyera, WWW 2004, New-York, USA, May 2004

2003

- "Device Independence for GUI, Voice and Multimodal Application: is it possible?", Stéphane Boyera, WWW 2003, Budapest, Hungary, May 2003
- "Les Travaux du W3C dans le domaine Device Independence" , Stéphane Boyera, Intech Seminar, Sophia Antipolis, February 2003 (french)

2002

- "Device Independent Authoring Workshop", Member of the Program committee.
- "Panel on Device Independence", Stéphane Boyera, WWW 2002, Hawai USA, May 2002

- "Device Independence Principles", Stéphane Boyera, Cross Media Publishing Workshop, Bohn, Germany, March 2002
- "Workshop on the Delivery Context", Stéphane Boyera, Organizer, Sophia-Antipolis, France, March 2002

Pre-2000 Scientific Publications

- "Un générateur de graphes de connaissance pour KATEMES", Stéphane Boyera, Rapport de DEA, Université de Nice, September 1993
- "Simuler le comportement macroscopique d'un modèle conceptuel", Stéphane Boyera, P.-A. Tourtier, O. Corby, , JAVA-94 (Journées Acquisition Validation et Apprentissage), Strasbourg, March 1994
- "Validating at early stages with a causal simulation tool", P.-A. Tourtier, Stéphane Boyera, EKAW-94, Hogaarden, 1994.
- "Etat de l'art sur la multi-expertise", S. Boyera, O. Corby, Deliverable L.3.2.2-1, Genie Project, June 1995